An Open, Site-Specific, Multi-Cell 5G Digital Twin: Ray-Traced Throughput, an OpenAirInterface-Grounded Link Model, and a Closed-Loop Traffic-Steering rApp

Author Name(s)
Affiliation
Organization
City, Country
email@domain

***Abstract*—Network digital twins are increasingly used to develop and validate O-RAN control applications before deployment, but open platforms trade off protocol-stack realism, site-specific propagation, network scale, and closed-loop control—none combines all four. We present an open-source, CPU-only digital twin that (i) builds a multi-cell, multi-user 5G scene from OpenStreetMap and ray-traces every cell-to-user channel with NVIDIA Sionna RT; (ii) computes per-user downlink throughput from a link model *measured on OpenAirInterface's (OAI) real physical layer*, using per-resource-element SINR with EESM link abstraction; (iii) hosts an O-RAN traffic-steering rApp; and (iv) grounds per-link realism by injecting the exact ray-traced channel into a live OAI gNB–UE link. Across three cities and eight random layouts each (24 runs), the rApp raises median per-user throughput by $9.5 \pm 11.3\%$ and served cell-edge throughput by $22.4 \pm 24.8\%$, improves Jain fairness by 9.1%, and reduces peak cell load by 0.75 users, while leaving aggregate rate and coverage-limited outage unchanged. A configuration sweep shows per-user throughput rising 1.7–$1.8\times$ as inter-cell load falls to idle. We outline the path to a fully integrated multi-UE closed loop, and release the platform, harness, and results as open source; everything runs on commodity CPU hardware.**

***Index Terms*—digital twin, 5G NR, OpenAirInterface, Sionna RT, ray tracing, O-RAN, rApp, traffic steering, link abstraction.**

I. Introduction

The digital twin is emerging as the environment in which AI/ML-driven RAN control (O-RAN rApps/xApps) is trained and validated before it touches a production network, and its value hinges on data fidelity. Open tooling splits into camps that do not meet: system-level simulators scale to many cells but abstract the physical (PHY) layer [8]; OAI-based emulators execute the real 5G-NR stack [6] but, with realistic propagation, cover a single link; commercial twins close the rApp loop but are proprietary.

The fidelity that matters for control-application development is not merely a faithful channel: it is the end-to-end mapping from site geometry to the KPIs a controller observes and acts on. A traffic-steering rApp, for example, decides which cell serves each user from signal, interference, and load; those quantities depend jointly on 3D propagation (site-specific), on the neighbouring cells (multi-cell), and on a link model that turns SINR into a real modulation-and-coding outcome (protocol-stack realism). Evaluating such a controller therefore requires all of these together, and—because random deployments vary widely—across many layouts rather than a single scenario. This motivates both our platform and our evaluation methodology.

Four capabilities matter *simultaneously* for a useful RAN twin—a real protocol stack, a site-specific ray-traced channel, a multi-cell network, and a closed control loop—and no open platform combines them. We contribute such a platform—integrating the four as components—with a statistically rigorous, honest evaluation: the multi-cell twin and its rApp run today, the real stack is grounded per link, and the remaining full-stack loop is designed end-to-end (Sec. VI-E). Our contributions are:

1) An open, CPU-only, multi-cell/multi-user twin whose throughput is grounded in OAI's measured PHY, driven by per-resource-element SINR and EESM.
2) A traffic-steering rApp evaluated over 24 runs across three cities, with robust statistics and explicit outage accounting.
3) Verified injection of the exact ray-traced channel into a live OAI link, plus a twin-to-OAI inter-cell-interference noise-floor bridge, and a design for the full-stack closed loop (multi-UE Standalone, FlexRIC KPM, and the rApp acting on OAI) that these components enable.
4) A reproducible experiment harness and results, released as open source.

II. Background

Link-to-system abstraction. System-level evaluation cannot afford full PHY processing per user, so it relies on link-to-system (L2S) mapping: a link-level simulator characterises the SINR-to-block-error-rate (BLER) relation for each MCS, and an effective-SINR mapping collapses a frequency-selective SINR vector into a single AWGN-equivalent value that indexes those curves. EESM is the widely used one-parameter form. The accuracy of L2S hinges on how faithful the underlying SINR-to-BLER curves are; we obtain ours from OpenAirInterface's real receiver rather than from idealised expressions.

O-RAN control loops. The O-RAN architecture externalises RAN control into applications: xApps on the near-real-time RIC (tens of ms, via E2) and rApps on the non-real-time RIC (seconds and above, via O1/A1). Traffic steering and load balancing are canonical rApp use cases, typically realised by adjusting per-cell handover offsets (cell-individual offsets, CIO). A digital twin lets such an rApp be trained and validated before it is allowed to touch a live network.

Twin fidelity. A twin used to rank or tune control policies must reproduce not only average link quality but the joint distribution of the KPIs a controller acts on, across the deployment randomness it will meet in the field. This makes multi-scene, multi-layout evaluation—and honest reporting of variance and outage—a fidelity requirement, not a nicety.

III. Related Work

Table I positions the closest open efforts on the four axes. Every existing open platform is missing at least one. OWDT [1] pairs OAI with Sionna RT for a faithful single gNB–UE link but is single-cell and monitor-only. Tiny-Twin [2] runs a real OAI stack for multiple UEs with a RIC loop, but single-cell from replayed channel traces (it lists inter-cell interference as future work). Orange's RIC-TaaP/ns-O-RAN and OpenTwin [3] close rApp loops across many cells—OpenTwin adds twin-to-real KPM self-correction—but on ns-3's abstracted PHY. Recent OAI+FlexRIC work [4] closes handover loops across real cells but over the air, without a site-specific ray-traced channel. Within OAI, an emerging ray-tracing channel emulator feeds the virtual-time simulator VRTSim (multi-node on its roadmap) and is a natural backend our injection step can adopt. The asterisk on our fourth axis is deliberate (Sec. VII): the four are integrated as *components* of one platform, with the rApp loop evaluated on the analytical twin and the real stack grounded per link; full rApp-driven multi-cell OAI is outlined as the next step in Sec. VI-E.

Commercial network digital twins close this gap in proprietary form: Rimedo Labs' NDT coupled with the VIAVI emulator trains and validates traffic-steering rApps before deployment, reporting double-digit throughput gains. Such tools demonstrate the industrial value of the closed loop but are neither open nor reproducible; our aim is an open, commodity-hardware counterpart whose every layer—geometry, propagation, link model, and control—can be inspected and re-run.

Two design choices distinguish our modelling from standard system-level practice [9]. First, the SINR-to-MCS lookup is measured from OAI's own PHY (`nr_dlsim`) rather than from idealised curves, so the network-scale KPIs inherit a real receiver's implementation losses. Second, our multi-cell interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [10]: instead of loading unused resources with a generic pattern, we derive each UE's interference from the ray-traced neighbour channels.

TABLE I
Open RAN digital-twin platforms on four axes.

Platform	Real stack	Site RT	Multi cell	Closed loop
OWDT [1]	✓	✓	×	×
Tiny-Twin [2]	✓	×[a]	×	✓
Orange ns-O-RAN / RIC-TaaP	×[b]	✓	✓	✓
OpenTwin [3]	×[b]	×	✓	✓
OAI+FlexRIC handover [4]	✓	×[c]	✓	✓
OAI VRTSim / RT emulator	✓	✓	×[d]	×
This work	✓	✓	✓	✓*

[a] replayed traces; [b] ns-3 abstract PHY; [c] over-the-air; [d] multi-node on roadmap; * components integrated, full loop in progress.

IV. System Architecture

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer that covers every cell and UE, and a real-stack layer that grounds sampled links in OAI. Both are driven by the same ray-traced channel, and the network data source is swappable so that synthetic and real deployments run through an identical path.

Geometry. A WGS84 bounding box drives an OpenStreetMap (Overpass) query; footprints are extruded into a 3D Mitsuba scene with ITU materials.

Network. Cells and UEs come from a swappable source: a seeded random generator (N sites $\times$ 3 sectors, uniform UE drops) or a CSV source for real coordinates (e.g., OpenCelliD).

Channel. Sionna RT [5] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE and computes the channel frequency response for every cell-to-UE link over an OFDM resource grid.

KPI engine. Let $\mathbf{H}_{u,c}(k) \in \mathbb{C}^{N_r \times N_t}$ be the ray-traced channel of link $c \to u$ on resource element (RE) k, and P_c^{tx} the cell transmit power. The wideband received power is $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \, \overline{\|\mathbf{H}_{u,c}\|_F^2}$, where the bar denotes averaging over antennas and REs. Each UE associates to the cell maximising received power biased by a cell-individual offset (CIO), the O-RAN control knob:

$$s(u) = \arg\max_c \; P_{u,c}^{\text{rx}} \cdot 10^{\,\text{CIO}_c/10}. \tag{1}$$

Because each cell serves one UE at a time under airtime sharing, the optimal single-stream precoder is the matched filter, so the serving per-RE gain is the largest singular value $\sigma_{\max}$ of the serving channel. With flat per-RE transmit power p_c, a neighbour load factor $\rho \in [0,1]$, and thermal noise $N_0 = kTB \cdot \text{NF}$, the per-RE SINR is

$$\gamma_u(k) = \frac{p_s \, \sigma_{\max}\big(\mathbf{H}_{u,s}(k)\big)^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \tag{2}$$

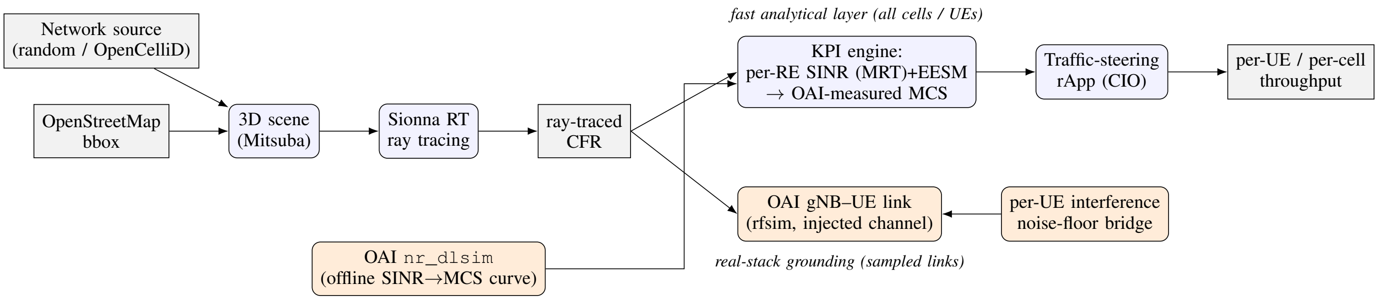

Fig. 1. Architecture. OpenStreetMap geometry and a swappable network source feed Sionna RT; the ray-traced channel drives (top) an analytical multi-cell KPI engine and the traffic-steering rApp, and (bottom) a live OAI link over the exact channel with a per-UE interference noise floor.

Algorithm 1 Proportional-fair traffic steering

1) CIO $\leftarrow \mathbf{0}$; evaluate R_u and U via Eqs. (1)–(4).
2) **repeat** (sweep over cells):
   a) for each cell c: line-search CIO$_c$ over a grid (others fixed) and keep the value maximising U.
3) **until** no cell's line-search improves U (local optimum).

The per-RE SINRs are compressed to one effective SINR by exponential effective SINR mapping (EESM), which preserves the frequency selectivity the ray tracer produces:

$$\gamma_u^{\mathrm{eff}} = -\beta \, \ln\Big(\tfrac{1}{K}\sum_k e^{-\gamma_u(k)/\beta}\Big), \tag{3}$$

with per-modulation β calibratable by a single scale factor. The effective SINR is mapped to a modulation-and-coding scheme (MCS) and spectral efficiency η_u through a curve Φ_{OAI} *measured offline from OAI's* `nr_dlsim` (real LDPC decoding and rate matching), i.e. $(m_u, \eta_u) = \Phi_{\mathrm{OAI}}(\gamma_u^{\mathrm{eff}})$. On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s}\,\eta_u\,(1 - \mathrm{BLER}_{\mathrm{tgt}}), \tag{4}$$

where B_s is the serving cell bandwidth and K_s its number of attached UEs.

Traffic-steering rApp. The rApp adjusts the CIO vector to maximise the proportional-fair utility $U(\mathrm{CIO}) = \sum_u \log R_u$ (Alg. 1). Because U is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus; we therefore line-search each cell's CIO over a grid (coordinate ascent), which is monotonically non-decreasing in U by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

Real-stack grounding. A bridge converts each link to time-domain taps injected into a live OAI gNB–UE link (a patch to OAI's channel module); a second bridge writes each UE's twin-computed inter-cell interference as a per-client OAI noise floor, so a single-link OAI run reflects its multi-cell context.

V. Experimental Setup

Scenes. Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry fetched/built once per scene and reused across seeds. **Random layouts.** Eight seeded layouts per scene—24 runs (323 s on one CPU server). **Configuration sweep.** Neighbour load factor in $\{0, 0.25, 0.5, 0.75, 1.0\}$ on a cached channel per scene. Radio and topology parameters are summarised in Table II. **Metrics.** We report per-UE downlink throughput (median and mean); the *served cell-edge* rate, defined as the 5th percentile among UEs with non-zero throughput (robust to coverage holes, unlike a raw 5th percentile which collapses to zero once outage exceeds 5%); the *outage rate*, the fraction of UEs with zero throughput; Jain's fairness index over per-UE rates; and the peak cell load (largest number of UEs on any cell). rApp gains are reported as mean ± standard deviation over the eight seeds per scene, and pooled across all 24 runs; a gain is undefined (excluded) for a run whose baseline metric is zero.

TABLE II
Radio and topology parameters.

Parameter	Value
Carrier frequency / bandwidth	3.5 GHz / 20 MHz
Subcarrier spacing	30 kHz (numerology 1)
Cell transmit power	46 dBm
BS array / UE antennas	4×1 / 1
Receiver noise figure	7 dB
Ray-tracing interaction depth	3
Cells / UEs per layout	9 / 30
Default inter-cell load ρ	1.0

VI. Results

A. Traffic-steering rApp

Across 24 runs, relative to strongest-cell association (Tables III–IV, Fig. 2), the rApp helps exactly the users a strongest-cell policy under-serves—improving the median, cell-edge, and fairness metrics—while shedding peak load. In absolute terms the overall median per-UE rate rises from 8.5 to 9.2 Mbps and the served cell-edge rate from 3.5 to 4.2 Mbps, whereas the mean is essentially unchanged (11.4 Mbps). Three honest observations follow: aggregate rate barely moves ($+0.5\%$; the proportional-fair objective trades sum for fairness), outage is unchanged (steering relocates covered users, it does not create coverage), and variance is high (layout-dependent)—motivating multi-seed evaluation. The controller

TABLE III
RAPP GAINS VS. STRONGEST-CELL (MEAN $\pm$ STD OVER SEEDS).

Metric	Overall	Berlin	Paris	Manh.
Median tput gain	$9.5 \pm 11.3\%$	10.5%	4.5%	13.3%
Served edge gain	$22.4 \pm 24.8\%$	41.0%	6.0%	20.2%
Jain fairness gain	$9.1 \pm 11.0\%$	11.2%	4.4%	11.5%
Peak load (UEs)	-0.75	-0.88	-0.50	-0.88
Mean tput gain	$0.5 \pm 5.8\%$	-0.7%	-0.3%	2.5%
Outage change	0.0 pts	0.0	0.0	0.0

TABLE IV
ABSOLUTE PER-UE THROUGHPUT, BASELINE$\rightarrow$RAPP (MBPS, MEAN OVER SEEDS).

Metric	Berlin	Paris	Manh.	Overall
Median tput	$8.3 \rightarrow 9.2$	$9.0 \rightarrow 9.4$	$8.2 \rightarrow 9.1$	$8.5 \rightarrow 9.2$
Served edge	$2.7 \rightarrow 3.9$	$3.8 \rightarrow 4.1$	$4.1 \rightarrow 4.7$	$3.5 \rightarrow 4.2$
Mean tput	$11.3 \rightarrow 11.2$	$12.2 \rightarrow 12.1$	$10.8 \rightarrow 10.9$	$11.4 \rightarrow 11.4$

itself is inexpensive: the coordinate ascent reaches a local optimum in a few per-cell line-search sweeps, keeping the loop well within an rApp's non-real-time (seconds-and-above) budget.

The per-scene breakdown is instructive (Fig. 3). Berlin, the densest scene, shows the largest served cell-edge gain ($+41\%$): dense blockage produces stronger load imbalance for the rApp to exploit. Paris shows the smallest gain ($+6\%$) and the highest outage (23.8%): when many users lack coverage entirely, steering has little covered load to rebalance. Manhattan sits in between. The consistent signal across all three cities is that the direction of the effect is robust—median throughput, cell-edge rate, and fairness improve while peak load falls—whereas the *magnitude* is strongly deployment-dependent, which is exactly the kind of conclusion a single-scenario study would misrepresent.

B. Coverage and outage

Mean UE outage is 16.2% at full load, and scene-dependent (Berlin 8.8%, Manhattan 16.3%, Paris 23.8%), reflecting random placement over the extent. For a fixed layout, outage is coverage-limited (path-loss-driven) and insensitive to inter-cell load, while served users are interference-limited (Fig. 4).

C. Sensitivity to inter-cell load

As the neighbour load factor falls from 1.0 to 0, mean per-UE throughput rises 1.7–1.8$\times$ (Berlin 13.6 $\rightarrow$ 22.7, Paris 14.9 $\rightarrow$ 25.2, Manhattan 11.9 $\rightarrow$ 22.0 Mbps; Fig. 5), confirming served users are interference-limited. Combined with the outage result, this separates two regimes the twin must represent distinctly: a coverage-limited tail, set by path loss and insensitive to load, that steering and load control cannot fix; and an interference-limited body, where both the neighbour load and the rApp's load-balancing act. A twin that reported only average throughput would blur these, and an rApp tuned on such a twin could over-claim gains that evaporate in the coverage-limited tail of a real deployment.

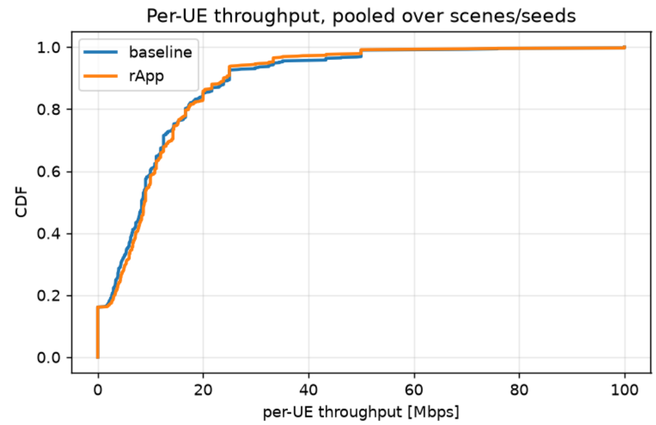

Fig. 2. Per-UE throughput CDF, baseline vs. rApp (pooled over scenes/seeds).

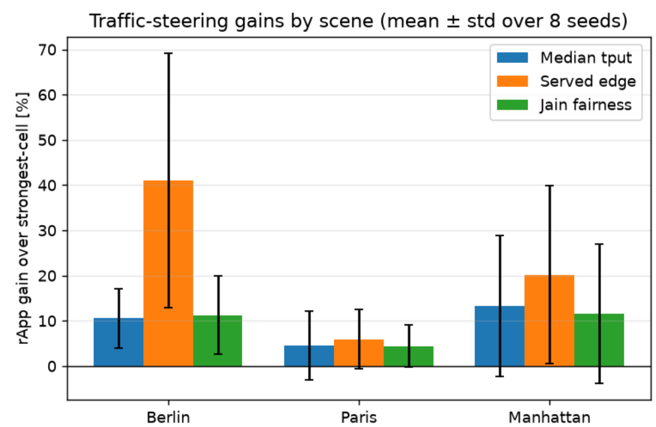

Fig. 3. rApp gains over strongest-cell association by scene (mean $\pm$ std over eight seeds). The direction is consistent; the magnitude is deployment-dependent, with the widest spread on the served cell-edge.

D. Real-stack grounding (single link)

The bridge converts a link's ray-traced channel frequency response to a sample-spaced impulse response (IFFT), retains the strongest causal taps, normalises them to unit energy so their shape carries the multipath while the overall gain is carried separately, and maps each tap delay to a sample index. These taps override the software-radio-simulator's channel through a small patch to OAI's channel module. A companion bridge expresses each UE's twin-computed inter-cell interference as a noise-floor rise, $\Delta N_u = 10\log_{10}\big((I_u + N_0)/N_0\big)$, written as a per-client OAI noise power, so a single-link run reflects its multi-cell context. OAI builds and runs on the same CPU host; a gNB–UE link reports real PHY KPIs (downlink SNR $\sim$15 dB, MCS 9), and injection is confirmed in the gNB log (`[RT] injected 4 taps ... channel_length=49`); the link then runs its PHY/MAC (HARQ, link adaptation) over the site-specific channel. Absolute path loss is masked by automatic gain control and uplink power control in phy-test mode, so the proof of grounding is

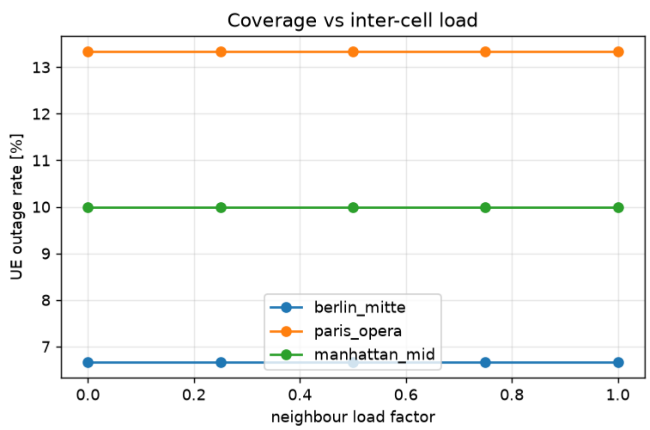

Fig. 4. UE outage vs. inter-cell load factor, for one cached layout per scene.

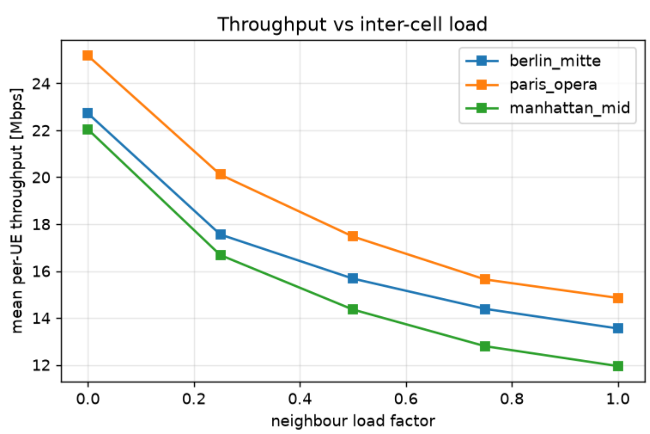

Fig. 5. Mean per-UE throughput vs. inter-cell load factor, for one cached layout per scene.

the injection and the multipath the receiver equalises, not an absolute level change; absolute goodput requires the full-stack Standalone path outlined in Sec. VI-E.

E. Toward full-stack integration

The verified components above compose into a full-stack closed loop, which we describe as the immediate next step. A multi-UE Standalone deployment (OAI 5G core in containers, gNB, and N UEs in network namespaces) would place each UE on its ray-traced channel and computed interference noise floor; FlexRIC [7] would stream per-UE KPMs (E2SM-KPM); iperf3 would supply real per-UE goodput; and the traffic-steering rApp would issue CIO/handover actions over O1/telnet, closing the loop on the real stack. The headline measurement this enables is the *fidelity gap*—the difference between the rApp gain the analytical twin predicts (Table III) and the gain the real stack delivers—arguably the most useful single number a network digital twin can report. The bridges of Sec. III (channel injection and per-UE interference noise floor) are the mechanism; the integration requires a host with Docker and root privileges.

F. Implications for twin-based rApp development

Two practical conclusions follow. First, a network digital twin intended to *rank* or *tune* control policies should report distributions over many random deployments, not a headline number from one scenario: our own single-seed pilot suggested a uniformly large gain, which the 24-run study tempered into a robust-direction/variable-magnitude result. Second, coverage and load are distinct problems—traffic steering improves the experience of *covered* users but cannot rescue users in outage—so a twin used to size an rApp's benefit must account for the coverage regime (here, via the outage rate and the load-sensitivity sweep) rather than reporting served-user throughput alone. These are precisely the kinds of design insights a twin is meant to surface before deployment.

G. Reproducibility and open release

All results are produced by an open experiment harness that fetches each scene's geometry once, reuses it across seeds (so only the random placement and the ray tracing re-run per layout), and aggregates per-run statistics with means, medians, and outage. A single command reproduces the 24-run matrix and the load sweep on a commodity CPU in about five minutes; each scene is ray-traced in roughly one minute. The twin, the OAI-measured link curve Φ_{OAI}, the traffic-steering rApp, and the OAI bridges (channel injection and per-UE interference noise floor) are released together, so that both the fast-layer network results and the per-link real-stack grounding can be re-run independently, and the full-stack integration of Sec. VI-E can be completed on a host with Docker and root.

VII. Discussion and Limitations

We state these plainly. (1) The rApp closed loop is currently evaluated on the multi-cell analytical twin; the OAI stack provides per-link ground truth and verified channel injection but is not yet *driven* by the rApp in one running system. Section VI-E outlines this integration; the two bridges (channel injection and interference noise floor) are the mechanism that connects them. (2) Inter-cell interference is modelled as (frequency-selective) noise power rather than physically combined waveforms; waveform-level combining across cells requires cluster-, FPGA-, or GPU-class channel emulators and is out of scope for a CPU platform. (3) The evaluation is a static full-buffer snapshot, all cells share one carrier, and only the downlink is modelled; mobility (via a virtual-time simulator), fractional frequency reuse, and uplink are natural extensions. (4) rApp benefits are high-variance and strongly layout-dependent, so we report full distributions rather than a single figure; practitioners should treat the mean as indicative and the spread as the operative uncertainty. (5) Traffic steering does not address coverage: users in outage cannot be rescued by re-association, only by densification or power/tilt control. (6) Validation is stack-versus-model; calibration against field measurements, and the use of real cell-site databases in place of random layouts, are the highest-value next steps.

VIII. Conclusion

We presented an open, CPU-only, site-specific, multi-cell 5G digital twin that grounds throughput in OAI's physical layer, hosts a traffic-steering rApp, and transmits a real OAI link over the exact ray-traced channel. Across three cities and eight random layouts each, the rApp improves median and cell-edge throughput and fairness and relieves peak load, without inflating aggregate rate or fixing coverage—an honest, reproducible result. The full-stack closed loop (Sec. VI-E) is the planned integration that will make it the first open platform to bring all four ingredients together on commodity hardware.

Future work. Completing the full-stack loop (multi-UE Standalone, FlexRIC KPM, and the rApp acting on OAI) will let us report the twin-predicted-vs-stack fidelity gap on a control task—arguably the most useful single number a RAN twin can provide. Beyond that, real cell-site databases in place of random layouts, mobility through a virtual-time simulator, uplink and multi-carrier support, and calibration against field measurements are the natural directions; each slots into the existing pipeline without changing its structure.

References

[1] T. Iye *et al.*, "Open Wireless Digital Twin: End-to-End 5G Mobility Emulation with OpenAirInterface and Ray Tracing," *IEEE Access*, 2025 (arXiv:2503.12177).

[2] A. Mamaghani *et al.*, "Tiny-Twin: A CPU-Native Full-stack Digital Twin for NextG Cellular Networks," arXiv:2601.08217, 2026.

[3] "OpenTwin: Digital Twin–Driven Closed-Loop KPM Inference and Control for Open RAN," arXiv:2605.24662, 2026.

[4] "Towards Intelligent Mobility Management: Customized Handover in 5G O-RAN Networks," ACM, 2025.

[5] J. Hoydis *et al.*, "Sionna RT: Differentiable Ray Tracing for Radio Propagation Modeling," arXiv:2303.11103.

[6] F. Kaltenberger *et al.*, "Driving Innovation in 6G Wireless Technologies: The OpenAirInterface Approach," *Computer Networks*, 2025.

[7] R. Schmidt, M. Irazabal, and N. Nikaein, "FlexRIC: An SDK for Next-Generation SD-RANs," in *Proc. ACM CoNEXT*, 2021.

[8] R. Pegurri *et al.*, "Toward Digital Network Twins: Integrating Sionna RT in ns-3 for 6G Multi-RAT Networks," INFOCOM WKSHPS, 2025 (arXiv:2501.00372).

[9] 3GPP TS 38.214, "NR; Physical layer procedures for data."

[10] 3GPP TS 38.106, "NR; Repeater radio transmission and reception" (OCNG).